REWARD ERRORS BECOME PERSISTENT STATE: A CONTROLLED WRITE-TIME MEMORY INTERVENTION IN SELF-IMPROVING AGENTS

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ABSTRACT

Self-improving agents turn terminal reward into persistent natural-language memory, so evaluator error can alter the state reused later. We isolate this channel in released ReasoningBank/WebArena Shopping trajectories. Holding each source trajectory byte-identical, flipping only the reward-conditioned reflection mode changes all four complete paired memories (mean token-set Jaccard distance 0.735); two additional failure-arm writes remain provider-incomplete. On eight disjoint trajectories, reward-mode divergence (0.700) exceeds stronger semantic-preserving same-mode rewording (0.595), with paired excess 0.105 and exact $p = 0.0078$. Reusing paired states changes the next structured action in 7/12 aligned probes. A same-support fixed-evidence 4×4 terminal replication after an initial non-pass completes 256 fresh rollouts and yields mean absolute success-rate difference 0.15625 ($p = 0.00074$); 8/16 cells are zero and 84.1% of centered signed-effect variation lies in source–future interaction. A fresh no-memory control shows omission matches both branches in eight cells, is closer to success-memory in six, and closer to failure-memory in two. A second GLM-5.3 writer reproduces write divergence on 4/4 sources (mean distance 0.737), but its 256-call terminal replication yields 0.140625 ($p = 0.00012$), below the preregistered 0.15 effect floor, with two direction reversals among six cells nonzero under both writers. Thus the write-time channel transfers more robustly than downstream effect magnitude or direction.

1 INTRODUCTION

Self-improving agents often treat the end of one episode as the beginning of the next learning update. In external-memory systems, this update can happen without gradients. The agent summarizes a completed trajectory and stores the resulting memory. Later tasks retrieve that memory as context. The terminal evaluator therefore does more than score behavior. When its output controls memory construction, the evaluator also controls persistent agent state.

Evaluator unreliability is already well documented: JudgeBench and RewardBench expose disagreement and failure modes in LLM judges and reward models (Tan et al., 2025; Lambert et al., 2024). The released ReasoningBank implementation makes the downstream state channel explicit (Ouyang et al., 2026): score one selects a success-reflection writer, while score zero selects a failure-reflection writer over the same trajectory. Reflexion likewise turns evaluative feedback into persistent verbal state (Shinn et al., 2023). Thus our novelty is not that judges can be wrong or that outcome-conditioned text can be stored; it is the matched write-time intervention created when an erroneous terminal bit changes persistent state while the source experience is held fixed. Figure 1 shows this write-and-reuse channel.

This channel sits upstream of recent memory-valuation results. Memory Reward Inflation shows that proxy reward can inflate erroneous memories during reuse (Asadolahi et al., 2026); RoMeRL studies trajectory-level utility updates and the memory-reward trap (Yang et al., 2026); and the agent-fragility study measures run variance, task-order sensitivity, and memory underspecification while deliberately using ground-truth reward in its main memory experiments (Ye et al., 2026). These works cover valuation, reuse, and downstream variance. Our residual object is earlier: whether reward error changes *what is written* from an identical experience, and whether this write-time state intervention propagates under matched future evidence.

A naive test is insufficient because the success and failure writers use different instructions by construction. Different prompts can produce different text even when reward semantics are irrelevant. We therefore treat generic prompt sensitivity as the strongest simple reduction. On eight fresh trajectories, semantic-preserving same-mode rewordings perturb prompt wording more than the released success/failure difference, yet reward-mode memory divergence remains larger: 0.700 versus 0.595, with paired excess 0.105 and exact $p = 0.0078$. Reusing the same fixed operation-slot taxonomy gives the same qualitative separation at the structural level: mean between-mode slot distance is 0.556 versus 0.247 within mode.

The underlying write intervention is equally direct. We hold each released trajectory byte-for-byte fixed and flip only the reward-conditioned reflection mode. Among the four requested pairs that complete in both writer conditions, every pair changes normalized memory content, memory-item titles, and operation-slot sets; mean token-set Jaccard distance is 0.735 and mean slot-set distance is 0.493. Two additional failure-label calls terminate at the provider response-state layer before assistant text exists, so every complete-pair statement is explicitly conditional on paired completion. The complete-pair effect occurs for source trajectories from both original-outcome strata, so it is not an artifact of drawing only failed episodes.

We then carry the intervention across the episode boundary. A fixed-state action probe changes the next structured browser action in 7 of 12 aligned paired rollouts. The decisive downstream test freezes all four complete source-memory pairs and all four future tasks before additional policy calls and executes eight fresh rollouts per memory condition in every cell. All 256 calls complete. Mean absolute terminal success-rate difference is 0.15625, and a 100,000-draw within-cell permutation test gives $p = 0.00074$. The effect is not uniform: 8/16 cells are zero, and a descriptive two-way decomposition leaves 84.1% of centered signed-effect variation in the source–future interaction residual, rather than a source-only or future-task-only main effect.

The resulting picture is a controlled write-and-reuse chain. Reward changes the memory-writing objective while the source experience stays fixed. The written state then reaches future decisions under matched evidence, and repeated memory-conditioned rollouts change terminal outcome distributions. The measured conditional effects also yield an explicit between-memory variance component under hypothetical stochastic reward-label corruption. In this frozen configuration, evaluator error therefore becomes a persistent-state consistency problem rather than remaining only an endpoint-scoring error; we do not elevate the plug-in variance calculation into a general empirical corruption law.

Contribution and closest-work boundary. Our contribution is one controlled causal chain rather than a new memory-learning algorithm. We hold the source trajectory byte-identical and use the released ReasoningBank success/failure writer branches as a fixed intervention surface, so the terminal reward bit selects writer mode without learning a new construction policy. A same-mode semantic rewording control challenges ordinary prompt sensitivity, and the resulting paired states are then carried across the episode boundary into fixed-evidence action probes and a fully crossed 256-rollout terminal test. Prior work already learns memory construction from downstream reward or finer process feedback; we therefore do *not* claim reward-driven memory construction itself, the success/failure writer, reward-model robustness, or memory-reward amplification as new. The residual claim is narrower: an erroneous terminal bit can select a different durable state from an otherwise identical trajectory, and that state difference can propagate to later behavior in this frozen ReasoningBank/WebArena configuration.

2 Related Work

Reward and judge reliability. JudgeBench and RewardBench establish that LLM judges and reward models can be unreliable before their outputs are used for learning (Tan et al., 2025; Lambert et al., 2024); Plan-RewardBench shows additional brittleness on tool-integrated trajectories (Wang et al., 2026). Our question begins one step later: what persistent state is created when such a terminal label selects a memory-writing objective?

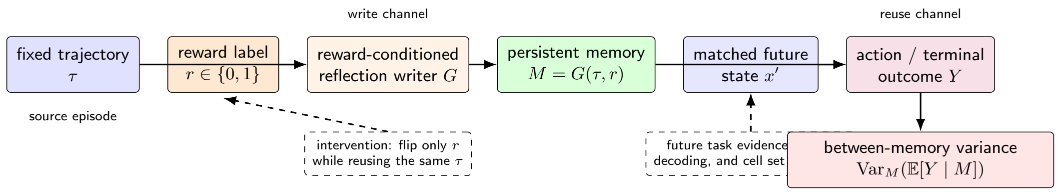

Figure 1: The reward-to-memory-to-outcome channel. The source trajectory is fixed while the reward label selects the reflection writer. The resulting persistent memory is then injected into a matched held-out task packet under a fixed policy configuration. Repeated outcomes identify the between-memory component of run variance.

Outcome-conditioned and learned memory construction. ReasoningBank explicitly distills different memories from successful and failed trajectories (Ouyang et al., 2026), Reflexion stores evaluative verbal feedback for later trials (Shinn et al., 2023), and Agent Workflow Memory extracts reusable procedures from interaction histories (Wang et al., 2024). More directly, Mem-α learns how to construct external memory from downstream question-answering reward (Wang et al., 2025), while AttriMem augments outcome reward with attribution-derived process feedback to learn a memory-construction policy (Li et al., 2026). Thus feedback-driven memory construction is already prior work. Our intervention does not learn a writer policy: it freezes an existing success/failure writer pair, holds the source trajectory byte-identical, and changes only the terminal label selecting that writer mode.

Memory valuation, reuse, and self-improvement variance. Memory Reward Inflation studies how imperfect proxy reward can be amplified as memories are scored and reused (Asadolahi et al., 2026). RoMeRL addresses misleading trajectory-level utility updates through reduced-order memory utility states (Yang et al., 2026). Ye et al. show run-to-run variance, curriculum sensitivity, and memory underspecification (Ye et al., 2026). These closest works cover downstream valuation and reuse; our residual object is upstream write-time state construction followed by matched downstream action and terminal tests.

Claim boundary. The paper therefore does not claim generic evaluator robustness, reward-driven memory construction as a new paradigm, universal harm from failure memories, or a new memory architecture. It identifies one narrower causal channel: with the construction policy held fixed, a reward-label error can select a different existing writer objective for an otherwise identical episode, and the resulting persistent state can change later behavior under the frozen configuration.

3 Problem Setup: Reward Error as a Persistent-State Intervention

Let task x produce trajectory τ, and let the terminal evaluator emit $r \in \{0, 1\}$. A reward-conditioned memory writer constructs

$$M(r) = G(\tau, r). \tag{1}$$

In the released ReasoningBank implementation, $r = 1$ selects the success-reflection objective and $r = 0$ selects the failure-reflection objective. The intervention studied here holds τ fixed and changes only this branch. The immediate causal estimand is therefore the paired persistent-state contrast $M(1)$ versus $M(0)$ for the same experience.

For a matched future task state x', let Y denote a downstream behavioral or terminal outcome under retrieved memory M. The reuse-stage contrast is

$$\Delta_Y(x') = \mathbb{E}[Y \mid do(M = M(0)), x'] - \mathbb{E}[Y \mid do(M = M(1)), x']. \tag{2}$$

The future task evidence and policy configuration are held fixed across the pair. This separates variation introduced by persistent memory state from variation in the observed future state.

3.1 VARIANCE AMPLIFICATION

Across self-improvement runs, reward errors can cause different memories to be written from otherwise comparable experiences. Let M denote the persistent memory state available to later tasks. The law of total variance gives

$$\mathrm{Var}(Y) = \mathbb{E}_M[\mathrm{Var}(Y \mid M)] + \mathrm{Var}_M(\mathbb{E}[Y \mid M]). \tag{3}$$

The second term is the component induced by variation in persistent memory state. If reward corruption changes which memory is written and the paired memories produce different conditional success probabilities, reward uncertainty contributes directly to this term. Section 7 estimates the paired conditional effect and converts it into an explicit corruption-rate variance curve.

This channel is immediate for external-memory agents. No parameter update is required: a mislabeled episode can change the context available to the very next retrieval.

4 EXPERIMENTAL SETUP

Frozen units and benchmark. We use released WebArena Shopping task metadata and released ReasoningBank trajectories rather than collecting a new self-improvement stream. F0 requests six source trajectories balanced across their original benchmark outcomes; only the four source pairs complete in both writer conditions and are eligible for paired or downstream analysis. The prompt-sensitivity control uses eight disjoint trajectories. Terminal confirmation crosses the four complete source-memory pairs with four future tasks frozen before terminal-policy calls, yielding 16 cells.

Models, controls, and metrics. DeepSeek-V4-Flash is the primary memory writer at temperature zero; Doubao Seed 2.0 Mini is the downstream terminal policy at temperature 0.2 with eight rollouts per memory condition and frozen cell. A source-independent no-memory control uses the same downstream configuration with 32 fresh calls total. A sequential GLM-5.3 replication first rewrites the same four source trajectories and unlocks the same 4-by-4 terminal protocol only after the frozen writer gate passes; its terminal gate remains mean absolute success-rate difference at least 0.15 with within-cell permutation $p < 0.05$. Primary observables are token-set Jaccard distance, structured-action disagreement, and terminal success-rate difference under the original deterministic WebArena evaluator. No-memory is a descriptive branch-location control, not a new global test.

Execution accounting and reliability. The study is inference-only, with no training or local GPU fine-tuning. Directly counted stages use 12 F0 writer requests (10 complete), 32 F0C writer requests, 96 F1D policy calls, 108 initial-terminal calls, 256 F2R1 calls, 64 O5 calls (32 execution-invalid plus 32 usable), eight repaired GLM writer calls, and 256 GLM terminal calls. The failed parent GLM writer attempt has zero scientific authority and an exact POST count that is unreconstructible after a concurrency race; its observable lower bound is nine. Hence 832 requests are exactly countable outside that failed parent and the full provider-POST lower bound is at least 841, excluding the unresolved low-level count of the earlier 12-unit action witness. Raw outputs are content-addressed and incomplete scientific units are never imputed.

5 F0: DOES THE REWARD BIT CHANGE WRITTEN MEMORY?

We begin with a paired intervention over released WebArena Shopping trajectories from the Salesforce fragility study. A deterministic sampler requests six source trajectories, balanced across original benchmark outcomes. For each trajectory, the task text and full action history are copied byte-for-byte into two ReasoningBank writer calls. One call uses the released success-reflection instruction and the paired call uses the released failure-reflection instruction. DeepSeek-V4-Flash writes memory at temperature zero, and raw provider responses are content-addressed before analysis.

Four of the six requested trajectory pairs return complete memory outputs in both conditions. The two incomplete requests terminate in the failure-label writer before assistant text is produced and are excluded from the paired effect calculation. Among the four complete pairs, every pair changes normalized memory content and the extracted memory-item title set. Token-set Jaccard distance is 0.691 for task 21 and 0.708 for task 22. It is 0.770 for both task 23 and task 25, giving mean distance

0.735. The 4/4 change rate is therefore an existence result conditional on paired completion. Panel (a) of Figure 2 shows the complete pair-level result.

The content changes are procedural as well as lexical. For task 21, success-conditioned memory centers on entering the review section and preserving direct quotations, whereas failure-conditioned memory emphasizes exhaustive pagination and extraction verification. For task 23, the success writer emphasizes evidence collection across pages while the failure writer emphasizes pagination state and broader synonym coverage. To make this distinction explicit without introducing another learned judge, we reuse a deterministic operation-slot taxonomy over the frozen memory-item titles: navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, verification/completeness, temporal filtering, and status filtering. The slot set changes in all four complete pairs; mean slot-set Jaccard distance is 0.493 (range 0.400–0.571). Moreover, every complete pair has a success-only slot in evidence capture, navigation, or query expansion, while verification/completeness appears as a failure-only slot in three of four pairs. This is a structural diagnostic rather than embedding-level semantic proof; Section 6 asks whether the same structure also changes under reward-preserving prompt rewording.

The F0 result establishes the first link of the mechanism: changing the reward-conditioned writer branch changes persistent memory under a fixed experience. Because the branch itself changes the writer instruction, the next section tests the strongest simple alternative explanation: generic sensitivity to prompt wording.

6 Ruling Out Generic Prompt Sensitivity

The reward intervention changes the reflection instruction together with the reward semantics. A generic prompt-sensitivity account therefore predicts that a sufficiently strong wording perturbation inside one reflection mode should produce comparable memory divergence. We test that reduction on eight released trajectories disjoint from F0 and balanced across original outcomes.

Before provider calls, we freeze semantic-preserving rewordings for both reflection modes. Under token-set Jaccard distance, the released success-versus-failure prompts differ by 0.114. The within-success rewording differs by 0.171 and the within-failure rewording by 0.152. The control thus perturbs surface wording more strongly while preserving the reflection objective.

Each fresh trajectory is written with the released success mode and the released failure mode. The same writer also processes one frozen semantic-preserving rewording of each mode at temperature zero. Let B_i be the memory distance between the two released reward modes and let W_i be the mean within-mode rewording distance. The frozen paired statistic is

$$\Delta_{\mathrm{prompt}} = \frac{1}{8}\sum_{i=1}^{8}(B_i - W_i). \tag{4}$$

All 32 control outputs complete. Mean reward-mode distance is 0.700, while the stronger same-mode wording control has mean 0.595. The paired excess is 0.105, and an exact one-sided sign-flip test gives $p = 0.0078$. This passes the frozen 0.10 effect margin together with the $p < 0.05$ gate. The reward-conditioned write effect therefore survives a same-information prompt-wording reduction that is deliberately stronger in lexical perturbation.

The same conclusion is visible at the procedural-structure level. Without changing the operation-slot taxonomy used in F0, we apply it to the frozen title sets from all eight control trajectories. Mean slot-set distance between the released success and failure modes is 0.556, compared with 0.247 for semantic-preserving rewordings within the same mode, an average structural excess of 0.309. Five tasks have positive excess, two tie, and one is negative. We treat this as a descriptive controlled diagnostic, not a second significance test or an embedding-semantic claim. Its role is narrower: the structural shift associated with reward mode is larger on average than the structural shift produced by reward-preserving wording changes under the same fixed taxonomy.

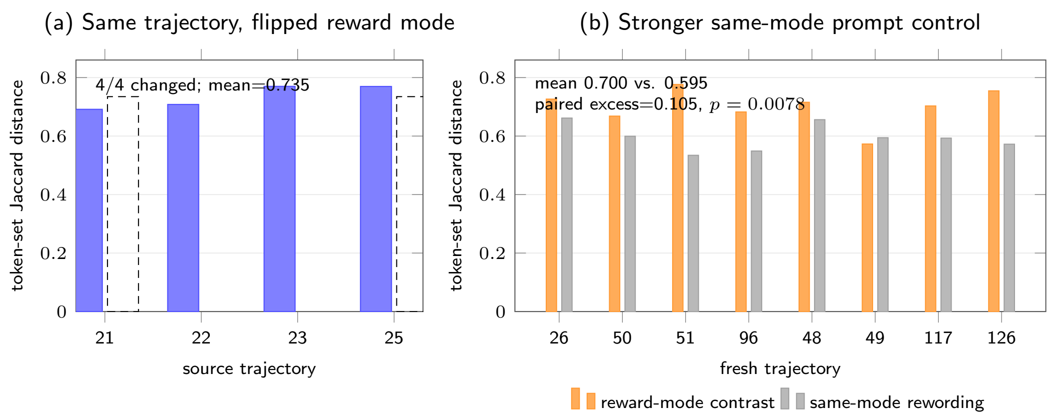

Figure 2: Write-channel evidence and its strongest simple control. (a) Among the four F0 pairs complete in both writer conditions, flipping only the reward-conditioned writer mode changes every memory. (b) On eight fresh control trajectories, reward-mode memory divergence exceeds stronger semantic-preserving same-mode prompt rewording. The control mean is 0.595 versus 0.700 between reward modes; paired excess is 0.105 with exact $p = 0.0078$.

7 C4: Does the Memory Perturbation Reach Future Behavior?

The write-channel result matters only if the persistent state difference survives reuse. We test this in two stages. The first asks whether paired memories can change an immediate decision under an identical browser observation. The second measures terminal success under a fully crossed source-memory by future-task design.

7.1 Fixed-state action reach

Four complete F0 pairs are mapped to four held-out Shopping states. Within each state, the future task prompt and browser observation are identical across memory conditions. The policy model and output schema are fixed. The original action-reach experiment contains 12 aligned success-memory versus failure-memory rollouts; 7 pairs choose different structured next actions, giving divergence rate 0.583 and passing the frozen existence falsifier. This establishes an action-level witness: paired memories can change the next decision under the same observation.

We separately stress-test the stronger claim that the first-action distributions are systematically separated across the frozen state set. This test uses eight rollouts per condition on every frozen state. Mean empirical total-variation distance between first-action distributions is 0.344. The 100,000-draw stratified permutation audit gives $p = 0.311$, so this stronger distribution-level test is a negative result: the available state set does not resolve systematic first-action distribution separation. The action-level evidence therefore rests on the paired existence witness (7/12 divergent next actions), not on a population-level distribution claim. Panel (a) of Figure 3 places this negative stress test beside the positive terminal evidence.

7.2 Fully crossed terminal confirmation

Terminal evaluation uses all four complete F0 source-memory pairs and four future tasks frozen before the first terminal-policy call. Every source pair is crossed with every future task, producing 16 cells. Within a cell, released browser evidence is byte-identical across memory conditions. Doubao Seed 2.0 Mini generates the terminal answer at temperature 0.2, and the original deterministic WebArena string-match evaluator scores success.

The initial frozen run uses three rollouts per memory condition. All 96 primary calls complete, with mean absolute success-rate difference 0.145833 and permutation $p = 0.160128$, so its dual gate does not pass. This non-pass was already committed as `730c2bc` before the F2R1 contract and before any F2R1 provider call. F2R1 is therefore a targeted *uniform replication after the initial non-pass*, not the first outcome-blind terminal experiment: it keeps the same four source memories, same four future tasks, same 0.15 effect floor, and same $p < 0.05$ gate, forbids source/task selection after outcomes, and increases only the fresh replication depth from three to eight per condition in every cell. All 256 F2R1 calls complete. Mean absolute success-rate difference is 0.15625 and the 100,000-draw within-cell permutation test gives $p = 0.00074$. Appendix 3 reports all 16 initial and confirmatory cells side by side. Eight confirmatory cells have zero observed contrast; six signed effects are positive and two negative. The two largest cells, 22/388 and 25/387, have signed effects 0.750 and 0.625 and account for 78.2% of squared-effect mass. We therefore interpret the result as finite distributional sensitivity, not a stable per-task directional effect. Because the two provider-incomplete source pairs are absent from this matrix, the result is conditional on the four completed source pairs. Panel (b) of Figure 3 shows the confirmatory signed matrix.

7.3 Fresh no-memory branch-location control

To locate the two confirmed memory branches relative to omission without changing the F2R1 gate, we freeze a source-independent no-memory control with eight fresh rollouts on each of the same four future tasks. All 32 recovery calls complete. No-memory success rates are 1.0, 0.0, 0.0, and 1.0 for future tasks 164, 385, 387, and 388. The same future-task baseline is shared across four source comparisons and is therefore not counted four times as independent evidence; we report Wilson intervals and cellwise Newcombe contrasts but no new global p-value. At the point estimate, omission matches both memory branches in eight cells, is closer to the success-memory branch in six, and closer to the failure-memory branch in two. For source/future 22/388, success-memory, failure-memory, and no-memory rates are 0.25, 1.0, and 1.0: the success-conditioned state is the branch that departs from omission. For 25/387, the corresponding rates are 0.125, 0.75, and 0.0, so both memories improve over omission but by different amounts. Thus the confirmed reward-conditioned contrast cannot be reduced to “failure memory is harmful” or “success memory is helpful”; which branch moves, and in which direction, is source–future specific.

7.4 Cross-writer generalization boundary

A sequential GLM-5.3 replication holds source trajectories, ReasoningBank reward-mode prompts, future-task support, downstream policy, evaluator, and terminal gate fixed. After one preregistered output-cap repair, all eight writer calls complete; all four pairs change content and title sets, with mean token-set Jaccard distance 0.737. The unchanged 4-by-4 terminal stage then completes 256/256 scorable units. Mean absolute success-rate difference is 0.140625 with $p = 0.00012$: the permutation criterion passes, but the effect misses the frozen 0.15 floor by 0.009375, so cross-writer terminal generalization is not established. Among six cells nonzero under both writers, four retain sign and two reverse (22/388: $+0.75 \rightarrow -0.25$; 23/387: $+0.125 \rightarrow -0.75$). Thus the write-time channel replicates more robustly than downstream effect magnitude or direction.

Where does the heterogeneity sit? We add a descriptive two-way decomposition of the centered signed 4-by-4 effect matrix, using only the already frozen F2R1 cells. Source-memory main effects account for 9.4% of centered sum-of-squares and future-task main effects for 6.5%; the remaining 84.1% is source–future interaction residual. The interaction is visible without modeling: source 21 is positive on future tasks 385/387 but negative on 388, while source 22 is positive on 385/388 but negative on 387. Conversely, future tasks 387 and 388 change sign across source memories. Thus the large effects are not well summarized as a fixed “bad memory” or a fixed “hard task” effect in this matrix. Outcome headroom also constrains what can be observed: future task 164 is at a joint 1.0/1.0 ceiling for every source pair and contributes zero squared-effect mass, whereas all observed squared-effect mass lies in the three non-ceiling future tasks. Write-stage divergence magnitude does not give a simpler explanation. Sources 23 and 25 have nearly identical token-set distances (0.770 in both cases) and identical operation-slot distance (0.500), yet their mean absolute terminal effects across the four future tasks are 0.031 and 0.156, respectively; source 22 has a smaller token distance (0.708) but the largest source-averaged effect (0.250). Thus “more different memory” is not a monotonic proxy for stronger downstream sensitivity in these four sources. Because task identity, workflow, binary-outcome headroom, and source–future compatibility are confounded in this finite matrix, these diagnostics characterize the observed heterogeneity rather than define a predictive transfer model.

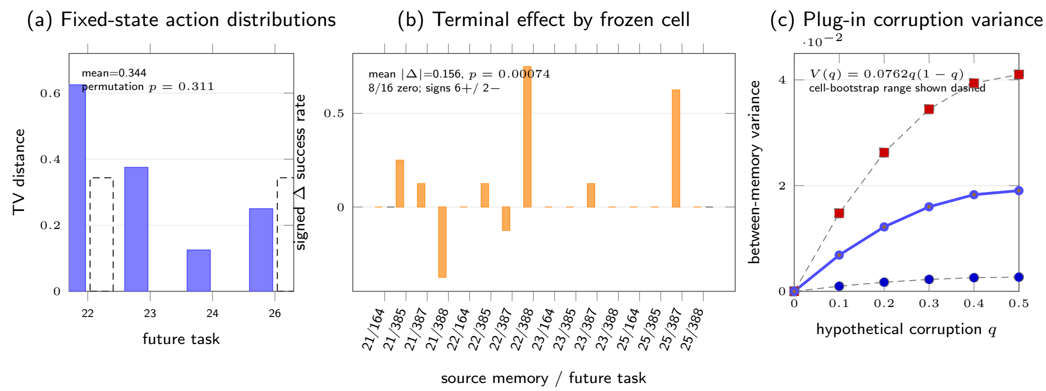

Figure 3: Downstream evidence for C4. (a) Across four frozen states with eight rollouts per memory condition, the larger first-action distributional stress test does not pass its permutation gate ($p = 0.311$). (b) The 16-cell matrix crosses four complete F0 source-memory pairs with four frozen future tasks; signed terminal effects reveal substantial heterogeneity, with aggregate mean absolute difference 0.15625 and $p = 0.00074$. (c) A plug-in decomposition gives the between-memory variance curve for hypothetical reward-label corruption probability q; dashed curves show the 95% cell-resampling sensitivity range.

The signed failure-memory minus success-memory effect averages 0.09375, while individual cells have both signs. Variance amplification therefore does not require every reward error to be uniformly harmful. It requires reward-conditioned memory state to alter conditional outcome distributions across runs.

7.5 Plug-in variance decomposition under reward corruption

Let $p_S(c)$ and $p_F(c)$ denote the confirmed success probabilities for success-conditioned and failure-conditioned memory in frozen cell c. If reward-label corruption selects the failure-conditioned memory with probability q, the between-memory success variance in that cell is

$$V_c(q) = q(1-q)\left[p_F(c) - p_S(c)\right]^2 . \tag{5}$$

This curve is a deterministic plug-in decomposition of the measured conditional effects, not an additional corruption-sweep rollout experiment. Across all 16 confirmatory cells, the mean squared conditional effect is 0.0762. Averaging over cells gives

$$\bar{V}(q) = 0.0762\, q(1-q). \tag{6}$$

A 100,000-repetition nonparametric bootstrap that resamples the 16 frozen cells gives a descriptive 95% sensitivity interval $[0.0107, 0.1641]$ for the mean squared effect. At $q = 0.5$, the plug-in estimate is 0.0190 with corresponding cell-resampling interval $[0.0027, 0.0410]$. The preregistered within-cell permutation test remains the primary confirmatory significance test. Figure 3(c) shows the plug-in curve together with this finite-cell sensitivity range.

8 Limitations

The F0 intervention requests six paired memories. Four pairs return complete outputs in both reward conditions. Both incomplete responses occur in the failure-label arm and terminate before assistant text is produced. Receipt-level forensics classify both as the same provider response-state error (`ArkResponseStateError`); neither failed call has a raw memory SHA, while the private provider response identifiers are archived and GET-only recovery returns no assistant text. By contrast, the four completed failure-label calls produce 777–1,296 output tokens. This evidence distinguishes the missing pairs from downstream memory-parser rejection, but it does not rule out a condition-dependent provider completion process. No memory text exists for those arms, so the direction of any resulting selection cannot be estimated. We therefore report the 4/4 write-channel result only among complete paired interventions and do not extrapolate it to all six requested trajectories.

The downstream action evidence has two strengths. The paired existence test records 7 divergent next actions among 12 aligned rollouts. The larger first-action distributional stress test yields mean TV 0.344 with permutation $p = 0.311$, so that stronger population-level action-distribution statement is not statistically resolved by the available state set. Terminal fixed-evidence evidence is stronger: the independently frozen 256-call replication passes its effect and permutation gates. These terminal experiments hold released browser evidence fixed. Source-faithful live browser navigation remains unexecuted because the released Shopping endpoint and reset service are unreachable from both execution hosts. This environment unavailability prevents the live extension and provides no counterevidence about the fixed-evidence effect.

The corruption-rate result is a variance decomposition of the confirmed paired conditional effects. A separate source audit shows that a full-bank corruption-mask sweep would not identify multi-memory interactions on the released ReasoningBank mechanism: retrieval embeds label-invariant task descriptions, uses top-1 with threshold 0.3, and injects only the single retrieved entry. Conditional on retrieval, only that source's corruption bit can affect the prompt; all other mask bits are inert. We therefore stop this sweep by matched simplification. A nontrivial corruption interaction would require multi-memory or content-dependent retrieval, which changes the substrate. Source-faithful retrieval-wrapper and live-browser transport remain separate replication scope.

GLM-5.3 changes all four paired memories and title sets, but its terminal replication yields 0.140625, below the preregistered 0.15 floor despite $p = 0.00012$, with two direction reversals among six cells nonzero under both writers. Thus the write-time channel has bounded cross-writer support, while writer-invariant downstream magnitude or direction does not. Additional writers and task domains are needed for broader claims. Token-set Jaccard is coarse, the operation-slot audit is not an embedding-based semantic analysis, and the 84.1% interaction share is descriptive rather than a predictive transfer model. The four source-independent no-memory baselines support branch location but not an independent $4 \times 4 \times 3$ factorial or a population-level memory-benefit claim.

9 Conclusion

Reward-conditioned memory writing creates a direct path from evaluator error to persistent agent state. Under an identical released trajectory, flipping the reward-conditioned reflection mode changes every complete paired memory, with mean token-set Jaccard distance 0.735. A disjoint eight-trajectory control shows that this separation exceeds stronger semantic-preserving same-mode prompt rewording by 0.105 on average, with exact $p = 0.0078$.

The persistent state difference reaches later behavior. A fixed-state probe changes the next structured action in 7 of 12 aligned pairs. The independently frozen fixed-evidence terminal replication completes 256 fresh rollouts over the full 4-by-4 source-memory/future-task matrix and finds mean absolute success-rate difference 0.15625 with permutation $p = 0.00074$. A fresh no-memory control rules out a uniform "success good / failure bad" interpretation of this contrast. GLM-5.3 then reproduces write-time divergence on all four sources, but its 256-call terminal replication reaches 0.140625 with $p = 0.00012$, missing the frozen 0.15 effect floor and showing two sign reversals among six cells nonzero under both writers. The evidence therefore supports a robust write-time channel more strongly than a writer-invariant downstream effect. A plug-in decomposition of the original confirmed conditional effects gives between-memory success variance $0.0762q(1-q)$ under reward-label corruption probability q.

In the released ReasoningBank/WebArena Shopping configuration studied here, the terminal evaluator is part of the state-update mechanism: its label selects persistent memory and the resulting memory changes later outcome distributions. Within this configuration, reward reliability is therefore a state-reliability requirement whenever outcomes determine what experience is written into persistent memory.

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A F0 Reproduction Details

A.1 Released trajectory sample

The F0 sample uses six task IDs from the released AWM shuffle-seed-42 Shopping parquet: 21 through 25, plus 47. The deterministic sampler selects the lowest task IDs with parseable trajectory records within each original-outcome stratum. Original failures are 21 through 22 plus 24. Original successes are 23 plus 25 and 47.

A.2 Memory construction intervention

Each released trajectory is summarized into an action trace. The same trace is inserted into two prompts. One prompt contains the released ReasoningBank success-reflection system instruction. The paired prompt contains the released failure-reflection system instruction. The task prompt stays fixed. The writer model stays fixed. Temperature stays at zero. The raw provider output is archived before metric computation.

A.3 Pair-level F0 results

Table 1: Complete paired interventions.

Task	Original outcome	Content changed	Token distance
21	failure	yes	0.691489
22	failure	yes	0.708108
23	success	yes	0.769953
25	success	yes	0.769608

The extracted memory-item title set changes for every complete pair. The mean token-set Jaccard distance is 0.734789. A deterministic post-ready strategy-slot audit maps the frozen titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification/completeness. All four complete pairs change their slot set. The slot-set Jaccard distances are 0.571, 0.400, 0.500, and 0.500 for tasks 21, 22, 23, and 25, respectively, with mean 0.493. This is a structural diagnostic over frozen titles, not an embedding-based semantic claim.

Provider failures for task 24 and task 47 are excluded from the paired analysis. Both are failure-label calls with `ArkResponseStateError`; neither produces assistant text or a raw memory SHA. Private response identifiers are retained and GET-only recovery finds no text. The four completed failure-label calls produce 777, 852, 1,113, and 1,296 output tokens. These receipts distinguish provider response-state failures from a later memory-parser rejection, while leaving arm-dependent completion bias unresolved.

B PROMPT-PERTURBATION CONTROL DETAILS

The control set is disjoint from the F0 requests. Each fresh trajectory is written under both released reward modes. One semantic-preserving rewording is also evaluated within each mode. Table 2 reports the paired distances used in the frozen sign-flip test.

Table 2: Full fresh prompt-perturbation control. Between is the released success-versus-failure memory distance; Within averages the two same-mode rewording distances.

Task	Original outcome	Between	Within	$B_i - W_i$
26	failure	0.726	0.662	0.065
50	failure	0.668	0.599	0.069
51	failure	0.776	0.534	0.242
96	failure	0.682	0.549	0.134
48	success	0.716	0.656	0.059
49	success	0.573	0.594	-0.022
117	success	0.703	0.593	0.110
126	success	0.755	0.572	0.182
Mean		0.700	0.595	0.105

C DOWNSTREAM CONFIRMATORY DETAILS

C.1 FIXED-STATE ACTION DISTRIBUTIONS

The distributional action stress test retains all four intervention states from the action-reach experiment and uses eight rollouts per memory condition. The per-state success-memory versus failure-memory TV distances are 0.625 for task 22 and 0.375 for task 23. They are 0.125 for task 24 and 0.250 for task 26. Their mean is 0.34375. The frozen 100,000-draw permutation audit gives $p = 0.310527$.

C.2 FROZEN TERMINAL MATRIX

The initial terminal experiment and F2R1 use the same complete F0 source set $\{21, 22, 23, 25\}$, the same future-task set $\{164, 385, 387, 388\}$, and the same aggregate absolute-effect statistic with the same dual gate: mean absolute difference at least 0.15 and within-cell permutation $p < 0.05$. The initial stage uses three fresh rollouts per condition per cell and is already frozen in Git commit `730c2bc` at 2026-08-22 00:01:42 +08:00; it yields 0.145833 with $p = 0.160128$ and does not pass. The separately authorized F2R1 contract is content-addressed as `ea9a2260...01f5`; its first provider receipt is generated at 2026-08-22 01:50:32 +08:00. It explicitly forbids source/future-task selection after outcomes and increases only uniform replication depth to eight fresh rollouts per condition in every cell. All 256 F2R1 calls complete, yielding 0.15625 with $p = 0.00074$. Thus F2R1 is a targeted same-support replication after an initial non-pass, not a claim that the confirmatory design was chosen before any terminal outcomes were observed.

C.3 FRESH NO-MEMORY BRANCH-LOCATION CONTROL

The no-memory control is frozen after F2R1 as a secondary reviewer-requested diagnostic. It uses the identical four future tasks, released evidence packets, Doubao Seed 2.0 Mini settings, evaluator,

Table 3: Initial F2 and F2R1 cell-level terminal rates on identical 4-by-4 support. S/F denote success-/failure-label memory.

Source	Future	F2 S	F2 F	F2 $\|\Delta\|$	F2R1 S	F2R1 F	F2R1 $\|\Delta\|$
21	164	1.000	1.000	0.000	1.000	1.000	0.000
21	385	0.000	0.333	0.333	0.000	0.250	0.250
21	387	0.000	0.333	0.333	0.125	0.250	0.125
21	388	1.000	0.333	0.667	1.000	0.625	0.375
22	164	1.000	1.000	0.000	1.000	1.000	0.000
22	385	0.000	0.667	0.667	0.000	0.125	0.125
22	387	0.000	0.000	0.000	0.125	0.000	0.125
22	388	1.000	1.000	0.000	0.250	1.000	0.750
23	164	1.000	1.000	0.000	1.000	1.000	0.000
23	385	0.000	0.000	0.000	0.000	0.000	0.000
23	387	0.000	0.000	0.000	0.125	0.250	0.125
23	388	0.667	1.000	0.333	1.000	1.000	0.000
25	164	1.000	1.000	0.000	1.000	1.000	0.000
25	385	0.000	0.000	0.000	0.000	0.000	0.000
25	387	0.000	0.000	0.000	0.125	0.750	0.625
25	388	1.000	1.000	0.000	1.000	1.000	0.000
Mean $\|\Delta\|$			0.145833			0.15625	

and eight-rollout depth, but no source-memory dimension. The earlier 12 exploratory no-memory calls are excluded. An initial 32-call execution is also excluded from science because a new local validator incorrectly required the provider's resolved model name to equal the requested alias; all 32 responses resolved to `doubao-seed-2-0-mini-260215`, which is exactly the versioned model name recorded in all 256 historical F2R1 receipts. After freezing that execution-layer diagnosis and archiving provider responses before local validation, the one-for-one recovery completes 32/32 calls. Both attempts are counted in execution cost; only the recovery outcomes enter the analysis.

Table 4: Fresh no-memory terminal baseline. Wilson intervals are per future task and are not replicated across source memories.

Future task	Successes	Rate	Wilson 95% interval
164	8/8	1.000	[0.676, 1.000]
385	0/8	0.000	[0.000, 0.324]
387	0/8	0.000	[0.000, 0.324]
388	8/8	1.000	[0.676, 1.000]

Combining these four source-independent baselines descriptively with the 16 F2R1 cells yields eight cells where omission matches both memory branches, six where omission is closer to the success-memory branch, and two where it is closer to the failure-memory branch. The largest illustrative contrasts are source/future 22/388, where $(p_S, p_F, p_0) = (0.25, 1.0, 1.0)$, and 25/387, where $(p_S, p_F, p_0) = (0.125, 0.75, 0.0)$. The first isolates the success-conditioned memory as the deviation from omission; the second places both memories above omission. Because each p_0 is shared by four source rows, we report cellwise score intervals and no new global significance test.

C.4 Cross-writer replication and frozen negative boundary

The O6 writer-family test is sequential. Stage 1 first rewrites only source tasks 21, 22, 23, and 25 under the same released success/failure ReasoningBank prompts using GLM-5.3 at temperature zero. The parent 2,200-token attempt is scientifically invalid after two failure-mode responses terminate at the exact output cap with no assistant text; an overlapping-runner race also makes its provider POST count reconstructible only as a lower bound of nine. The single preregistered repair changes only the output cap to 4,096 and adds a single-writer transaction lock. Its eight calls all complete; all four pairs change content and title sets, with token-set distances 0.783, 0.728, 0.715, and 0.723 (mean 0.737482).

Stage 2 keeps the original Doubao Seed 2.0 Mini downstream policy, four future tasks, evidence packets, evaluator, eight-rollout cell depth, and dual terminal gate unchanged. All 256 cells-by-condition rollouts are scorable. The observed mean absolute effect is 0.140625 and the 100,000-draw permutation test gives $p = 0.00012$. The statistical gate passes but the frozen 0.15 minimum-effect gate does not, so the registered cross-writer terminal generalization claim is not established. Table 5 shows that the difference is not a uniform attenuation.

Table 5: Signed failure-minus-success terminal effects under the original DeepSeek-V4-Flash writer (D) and the GLM-5.3 writer (G).

Source	Future	D effect	G effect
21	164	0.000	0.000
21	385	0.250	0.250
21	387	0.125	0.375
21	388	-0.375	0.000
22	164	0.000	0.000
22	385	0.125	0.000
22	387	-0.125	-0.125
22	388	0.750	-0.250
23	164	0.000	0.000
23	385	0.000	0.000
23	387	0.125	-0.750
23	388	0.000	0.000
25	164	0.000	0.000
25	385	0.000	0.000
25	387	0.625	0.500
25	388	0.000	0.000

Among the six cells that are nonzero under both writers, four retain direction and two reverse. Reusing the already frozen writer-independent no-memory rates only as descriptive baselines yields 10 GLM cells equidistant from omission, four closer to the success-memory branch, and two closer to the failure-memory branch; no additional provider calls or global test are introduced. This supports the upstream write-channel replication while preserving the failed terminal-invariance gate.

C.5 Heterogeneity and corruption-rate decomposition

For each frozen cell, the confirmatory run estimates paired conditional success probabilities. Eight of the 16 cells have zero observed success-rate contrast. Six have positive failure-memory minus success-memory effects and two have negative effects. The largest squared effects occur at source/future pairs 22/388 and 25/387; together they contribute 78.2051% of the total squared-effect mass. This sign and magnitude heterogeneity is why the paper treats the result as a variance channel instead of a uniform directional harm effect.

If reward-label corruption selects the failure-conditioned memory with probability q, the induced between-memory variance equals $q(1-q)\Delta_c^2$ in cell c. The exact mean squared conditional effect across all 16 cells is 0.076172. Averaging over cells gives the curve $0.076172q(1-q)$, with value 0.006855 at $q = 0.1$, value 0.014282 at $q = 0.25$, and value 0.019043 at $q = 0.5$.

For finite-cell sensitivity, we run a fixed-seed nonparametric bootstrap with 100,000 repetitions, resampling 16 source–future cells with replacement. The percentile 95% interval for mean absolute effect is $[0.054688, 0.273438]$. The corresponding interval for mean squared effect is $[0.010742, 0.164062]$, which maps to $[0.002686, 0.041016]$ for $V(0.5)$. This bootstrap is descriptive sensitivity to the observed cell composition. The preregistered within-cell permutation test on the 256 confirmatory rollouts supplies the primary significance result.

C.6 Post-ready structural diagnostic

The Stanford-targeted diagnostic reuses the exact operation-slot taxonomy already frozen for the F0 title audit; it does not add a learned semantic judge. Applied to the eight disjoint F0C trajectories, the mean slot-set distance between released reward modes is 0.555655, whereas the mean distance induced by semantic-preserving rewording within the same mode is 0.246578. The mean between-minus-within structural excess is 0.309077; five tasks are positive, two tie, and one is negative. This is descriptive supporting evidence rather than a second hypothesis test. On the four complete F0 pairs, success-only operation slots include at least one of evidence capture, navigation, or query expansion in all four pairs, while verification/completeness is failure-only in three pairs. Provider receipts also show longer failure-mode responses in all four complete pairs (mean output-token difference +266.5; mean reasoning-token difference +268.0), which we report only as response-structure asymmetry, not as semantic quality.

C.7 Post-ready source–future interaction diagnostic

A two-way additive decomposition of the centered signed F2R1 cell effects is shown below. It is a finite-matrix description, not inferential ANOVA.

Table 6: Descriptive decomposition of centered signed terminal effects.

Component	Sum of squares	Share
Source-memory main effect	0.101562	9.4%
Future-task main effect	0.070312	6.5%
Source $\times$ future interaction residual	0.906250	84.1%
Total	1.078125	100%

The source and future marginals therefore summarize little of the signed effect structure. Source 21 is positive on future tasks 385 and 387 but negative on 388; source 22 is positive on 385 and 388 but negative on 387. Future tasks 387 and 388 likewise change sign across source memories. Table 7 adds benchmark-observable future-task attributes. Future task 164 is at a joint 1.0/1.0 terminal ceiling for every source pair, so it has no observable binary-outcome headroom and contributes zero effect mass. The other three tasks contain all observed squared-effect mass. With only four future tasks, start context, task identity, and outcome headroom are confounded; these rows characterize where the frozen effect occurred but do not define a general transfer predictor.

Table 7: Future-task attributes and concentration of the frozen terminal effect. "Home" means the task starts at the Shopping root rather than a product page.

Task	Start	Ref. items	Mean $\lvert\Delta\rvert$	Squared-mass share
164	Product	2	0.00000	0.0%
385	Home	8	0.09375	6.4%
387	Home	6	0.25000	35.9%
388	Home	2	0.28125	57.7%

A second descriptive check asks whether sources whose paired memories differ more at write time simply create larger downstream effects. Table 8 shows that this reduction is not supported by the four completed sources. Sources 23 and 25 are effectively matched on both reported write-divergence magnitudes but differ fivefold in source-averaged terminal sensitivity. Across the four points, the descriptive Pearson correlations are -0.730 for token distance and -0.367 for slot distance versus mean terminal $|\Delta|$. With $n = 4$ these correlations have no inferential status; they are included only to reject the simple monotonic reading that a larger textual or operation-slot perturbation necessarily produces a larger downstream effect.

Table 8: Write-stage divergence versus source-averaged terminal sensitivity. The final column averages $|\Delta|$ over the four frozen future tasks.

Source	Token distance	Slot distance	Mean terminal $\lvert\Delta\rvert$
21	0.691	0.571	0.188
22	0.708	0.400	0.250
23	0.770	0.500	0.031
25	0.770	0.500	0.156